BSc 1st semester Materials

Economic Botany and Ethnomedicine

Botany major

(ABN seal college)

created by -- Bikram Singha

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All scientific names included in theory syllabus

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(*) mistakes in test examination. → Tobacco

	Common name	Scientific name	Family
1)	Rice	Oryza sativa L.	Poaceae (Graminae)
2)	Wheat	Triticum aestivum L.	Poaceae (Graminae)
3)	Sorghum (Jowar)	Sorghum vulgare Pers.	Poaceae (Graminae)
4)	Pearl millet (Bajra)	Pennisetum typhoides L.	Poaceae (Graminae)
5)	Finger millet (Ragi)	Eleusine coracana L.	Poaceae (Graminae)
* 6)	Chick pea	Cicer arietinum	Fabaceae (leguminosae)
* 7)	~~Pegion~~ pea Pigeon	Cajanus cajan	Fabaceae (leguminosae)
8)	Cinchona	Cinchona officinalis	Rubiaceae
9)	Digitalis / Foxglove	Digitalis purpurea	Scrophulariaceae
10)	Papaver / opium poppy	Papaver somniferum	Papaveraceae
11)	Cannabis / Ganja	Cannabis sativa	Cannabaceae
(*) 12)	Tobacco	Nicotiana tabacum	Solanaceae
13)	Potato	Solanum tuberosum	Solanaceae
14)	Sugercane	Saccharum officinarum	Poaceae (Graminae)
15)	Tea	Camellia sinensis L.	Theaceae
16)	Coffee	Coffea arabica L.	Rubiaceae
17)	Okra	Abelmoschus esculentus	Malvaceae
18)	Cabbage	Brassica oleracea	Brassicaceae
19)	Mango	Mangifera indica L.	Anacardiaceae
20)	Orange	Citrus reticulata	Rutaceae
21)	Marigold	Tagetes erecta	Asteraceae
22)	Tuberose	Polianthes tuberosa	Asparagaceae
23)	Groundnut	Arachis hypogea (gaaa)	Fabaceae
24)	Coconut	Cocos nucifera	Arecaceae
25)	Soybean	Glycine max	Fabaceae
26)	Mustard	Brassica campestris	Brassicaceae
27)	Sal	Shorea robusta	Dipterocarpaceae
28)	Teak	Tectona grandis	Verbenaceae
29)	Jute	Corchorus capsularis olitorius	Tiliaceae

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Scientific name	Common name	Family
① Eclipta prostrata	Kesuti / Kalo kesuti / Bringraj	Asteraceae
② Sesbania grandiflora	Bokful / Humming bird tree	Fabaceae
③ Glycosmis arborea	Dantan / Ginberry	Rutaceae
④ Vitex negundo	Nishinda / ~~Guest~~ Chaste tree	Lamiaceae
⑤ Coccinia grandis	Telakucha / Ivy gourd	Cucurbitaceae
⑥ Alstonia scholaris	Chhatim / Milkwood / Devils tree	Apocynaceae
⑦ Artemisia vulgaris	~~Tet~~ Titepati / Nagdana /	Asteraceae
⑧ Andrographis paniculata	kalmegh / kalomegh /	Acanthaceae
⑨ Centella asiatica	Thankuni	Aplaceae
⑩ Rauvolfia serpentina	Sarpagandha / Indian snake root	Apocynaceae
⑪ Oroxylum indicum	Sunimala	Bignoniaceae
⑫ Terminalia chebula	Hortoki / Horitoki	Combretaceae
⑬ Terminalia bellirica	Boel	Combretaceae
⑭ Terminalia arjuna	Arjun	Combretaceae
⑮ Ocimum sanctum	Tulshi / Holy ~~Brasil~~ Basil	Lamiaceae
⑯ Justicia adhatoda	Basak	Acanthaceae
Para Rubber	Hevea brasilliensis X brasiliensis	Euphorbiaceae

{ coffee & cinchona → Rubiaceae
{ Dantan / Gin berry → Rutaceae

① Gamari → Gmelina arborea → Lamiaceae
② Shimul → Bombax ceiba → Malvaceae
③ Mahogany → Swietenia mahagoni → Maliaceae
④ Lemon → Citrus limon
⑤ lime → citrus aurantiifolia
⑥ grapefruit → citrus paradisi
⑦ Sweet orange → citrus sinensis
→ Rutaceae
⑧ Citronella → Cymbopogon nardus → Poaceae
⑨ Eucalyptus → Eucalyptus globulus → Myrtaceae

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ETHNOMEDICINE

scientific name

common name

family

parts used

uses

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Ethonomedicine

Scientific name- Eclipta ~~prostata~~ prostrata.
common name- Kesuti / Kalo Kesuti / Bringraj. / False daisy.
family name- Asteraceae.
parts used- leaves, stems, ~~roots~~ roots and the whole plant
Uses-
① Skin disease
② ~~Liver~~ traditional medicine for liver protection.
③ Hair growth.
④ Many cosmetics

Phytochemicals-
Eclalvasaponin, Tuleonin

Scientific name- Sesbania grandiflora
common name- Bokful / Humming bird tree.
family name- Fabaceae
parts used- ~~Ro~~ leaves, flower, pods, bark, roots, seeds, (whole plant)
Uses-
① Immune booster
② Fever, cough & cold
③ Diarrhoea

Phytochemicals-
> Alkaloids, flavonoids

Scientific name- Glycosmis arborea
Common name- Gin berry / Dantan Gach.
family name- Rutaceae
parts used- leaves & roots
Uses-
① leaves- juice- fever; worms, skin issues, bath for pain.
② Roots - anemia, ~~inflama~~ inflammation.

Phytochemicals-
> Flavonoids, Glycoborenine

Scientific name- Vitex negundo.
Common name- chaste tree / Nishinda
family name- Lamiaceae
parts used- Bark, leaves
Uses-
① Bark- Skin disease, cough, pain in chest
② leaves- Inflammation, cough, wounds.

Phytochemicals-
>Vitexin, Isovitexin

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Scientific name - Coccinia grandis
Common name - Ivy gourd / Telakucha
Family name - Cucurbitaceae
Parts used - leaves, fruits, Roots, stems.
Uses -
① leaves → for diabetis, anti-inflammatory, antioxident
② fruits → fever, asthama.
③ Roots → Diabetis, Joint ~~pant~~ ~~pain~~ pain.
④ Stems → Respiratory disease
Phytochemical
> Flavonoids, cucurbitacins

Scientific name - Alstonia scholaris.
Common name - Chhatim / Milkwood / Devils tree
Family name - Apocynaceae
Parts used - Stem bark, leaves
Uses -
① Stem bark - Fever, malaria, bronchitis, diarrhea, dysentery
② Leaves -

Phytochemical
> Flavonoids, Glycosides, Alkaloids

Scientific name - Artemisia vulgaris.
Common name - Nagdana / Titepati / Common mugwort
Family name - Asteraceae.
Parts used - leaf
Uses -
① GIT disease, menstrul problem, fever, neurological problem, skin disease.
Phytochemical
> Caffein acid, [illegible]

Scientific name - Andrographis periculata.
Common name - Kalmegh / Bitter wood
Family name - Acanthaceae
parts used - leaf
Uses -
① Cough & cold, immune booster, Diabetis, Blood purifier, malaria.
Phytochemical -
> [illegible]
Andrographolide, Flavonoids

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Scientific name- Centella asiatica
Common name- Thankuni /
Family name- Apiaceae
parts used- leaf
Uses-
① memory enhancer, wound healing, skin disease, anxiety reducer, stomach disease.

Phytochemical-
>Asiaticoside, Asiatic acid

Scientific name- ~~Rauvof~~ Rauvolfia serpentina
Common name- Sarpagandha / Indian snake
Family name- Apocynaceae
parts used- Roots
uses-

① Roots → Reserpine → ~~Hit~~ Hypertension, Anxiety & mental disorder, diarrhia, fever, malaria.

Phytochemical-
> Reserpine, Flavonoids (less), Alkaloids (more)

Scientific name- Oroxylum indicum
common name- Surimala /
Family name- Bignoniaceae
parts used- Bark, flower
Uses-
① Gynocological disorder, Arthrritis, diabetis, Respiratory disorder, digestive problem
Phytochemical- Flavonoids

Scientific name- Terminalia chebula
Common name- Hortoki / Horitoki
Family name- Combretaceae
parts used- Fruit
Uses-
① Cough & cold, Indigestion, Arthritis

Phytochemical
> Chebulic acid, gallic acid

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Scientific name- Terminalia bellirica
Common name- Bohera/
family name- combretaceae
parts used- fruit
Uses-
① Cough, asthma, pain relif, liver disease, eye disease

Phytochemical-
> gallic acid,

Scientific name- Terminalia arjuna
Common name- Arjun
Family name- Combretaceae
Parts used - bark
Uses-
① Heart disease, Hypertension, Blood circulatory problems.

Phytochemicals- Arjunolic acid, Terminic acid

scientific name- Ocimum sanctum
Common name- Holy ~~bf~~ basil / Tulshi
Family name- Lamiaceae
parts used- leaf
Uses-
① Cough & cold, Asthama, Bronchitis, Diabetis, skin disease, eye problem, dental problem.

Phytochemical- Flavonoids, Alkaloids.

scientific name- ~~Justica~~ Justicia adhatoda
Common name- Basak / Malabar nut
Family name- Acanthaceae
Parts used- leaf
Uses-
① Cough & cold, Asthama, Bronchitis, Throat infection, Fever, tuberculosis, malaria.

Phytochemical
> Alkaloids, Flavonoids.

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SPICES

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Spices

Common name- Fennel (Mourie)
Scientific name- Foeniculum vulgare Mil.
Family name- Apiaceae (~~Umbels~~ Umbelliferae)
parts used - Young leaves, seeds and fruit
Uses -
i) Used as flavouring agent.
ii) Use in curry.
iii) Flavouring agent for pickle.
iv) Used for extraction of essential oil.
v) Medicinally use for digestive issues.

Common name - Large ~~cad~~ cardamom (Bengal cardamom, bara elach)
Scientific name - Amomum subulatum Roxb.
family name - Zingiberaceae
Parts used - Fruit and seeds
Uses -
i) Large cardamon is widely used as a spice to add flavour and aroma to dishes such as biryani, curries, sweets etc.
ii) The seeds and pods are used in tea blends and spice mixes (like garam masala) for their pleasent fragnance.
iii) It is used in Ayurvedic medicine to aid digestion.
iv) Large cardamom oil is used in perfume and pharmaceutical industries.
v) The essential oil extracted from the seeds is also used in flavouring and aromatherapy products.

Common name - clove
Scientific name - Syzygium aromaticum L.
Family name - Myrtaceae
parts used - Dried flowers bud
Uses -
i) Used in dishes like biryanis, curries to add aroma.
ii) Due to its strong fragnance, clove is used in perfumes, air fresheners etc.
iii) Used in home remedies for treating cold, nausea & bloating.
iv) Improves digestion and boost immunity.
v) Clove oil is used in pharmaceuticals, cosmetics, toothpaste.

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Common name- Black peper. (kali mirch / Gol morich)
Scientific name- Piper nigrum
Family name - Piperaceae
parts used - Small one seeded fruits (Berries)
Uses -
i) It is use for its strong aroma that enhances the flavour of foods and beverages.
ii) A mix of black ~~pow~~ peper with honey use ~~for~~ as a home ramedy for cough, sore throat and respiratory issues.
iii) Black peper extract ~~oil~~ & oil are used in pharmaceutical, perfumary and food industries for flavouring, preserving and medicinal formulations.
iv) Black peper helps in digestion and improves metabolism.
v) It has antioxidant and anti inflamatory properties.

Common name - True cardamom (Chhota elaichi)
Scientific name- Elettaria cardamomum Maton.
family name- Zingiberaceae
parts used - Ripe fruits containing seeds.
Uses -
i) Use as flavaring agent in cooking.
ii) As of its strong and pleasent fragnance ~~&~~ it is used in tea blend
iii) Cardamom is valued in Ayurvedic and traditional medicine.
iv) Used as home ~~rem~~ remedy for cough and cold.
v) Cardamom oil and essence are used in food, cosmetics and pharmaceutical industries for flavouring, scenting & medical formulations.

Common name - Cinnamon
Scientific name- Cinnamomum zeylanicum Bl.
family name - Lauraceae
parts used - Bark of the stem.
Uses -
i)

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Common name- Turmeric (Haldi)
Scientific name- Curcuma longa L.
family name- zingiberaceae
parts used- Rhizome
Uses-
i)

Common name- Ginger (Ada)
Scientific name- Zingiber officinale
family name- zingiberaceae
parts used- Rhizome.
Uses -

Common name- Asafoetida (Hing)
Scientific name - Ferula foetida
Family name - Apiaceae (Umbelliferae)
parts used - Gum resins of the roots.
Uses -

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Common name- Tejpata
Scientific name- Cinnamomum tamala.
family name- Lauraceae
parts used - leaves.
Uses-

Common name- Coriender (Dhone)
Family name- Coriandrum sativum L.
Family name- Apiaceae (Umbelliferae)
parts used- Young leaves and mature seeds.
Uses-

Common name- cumin (Jira)
Scientific name- Cuminum cyminum L.
Family name- Apiaceae (Umbelliferae)
parts used- Fruits.
Uses-

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Common name- Khair
scientific name- Areca catechu
family name- Mimosae (Leguminosae)
parts used- The resinous extract obtained by boiling of Heard wood.
uses -

Common name- Saffron
Scientific name- Crocus sativus L.
Family name- Iridaceae
parts used- Stigma of the flower.
Uses -

Common name- Fenugreek (methi)
Scientific name- Trigonella foenumgraecum L.
Family name- Fabacee
parts used- Seeds.
uses -

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Common name- Mint (Pudina)
Scientific name- Metha piperita L.
Family name- Lamiaceae
Parts used- leaves.
Uses-

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Cultivation methods

okra
cabbage
marigold
tuberose

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vegetable

① Common name- Okra / lady's finger / Bhindi
Scientific name- Abelmoschus esculentus.
Family name- Malvaceae
part use- Fruits

origin
↓
Tropical Africa

Uses-

① Use as vegetable in soup, curries, fries, sambaretc.
② Agriculturilly use as short duration cash crop.
③ Medicinally use for controlling blood suger, improves immunity, rich in vit A, B & C.

Cultivated varieties- (In North Bengal)
①

Cultivation method-

① For okra cultivation climatic condition require hot(25°-35°c) humid, summer to rainy season.
② The soil pH requires (6-6.5) slightly acidic, sandy and well drained soil preferred for okra cultivation.
③ The seeds of okra soaked in-to water for 12-24 hours for germination.
④ After germinating the seedling are ~~placed~~ planted in a spacing of - 30-60 cm (for tall varieties)
- 20-40 cm (for short varieties).
⑤ for fertilizer NPK ada ratio of (40:20:20) used manure ~~with~~ ~~out~~ 10 tons/acar.

During harvesting ~~long~~ ~~and~~ 3-4 inch long & tender pods (45-60 days old) are harvested & marketed.

Pest & Disease

① Yellow vein mossaic disease → cause by → yellow vein mossaic virus → Transmitted by → white flys.
② Also some insect larva are common pest in the okra cultivation.



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Common name- Cabbage
Scientific name- Brassica oleracea
Family name- Brassicaceae
parts used- Young leaves and stems.
Uses -

① Used in curries, soups and salad.
② Also used in pickels.
③ Medicinally used as Antioxidant & rich in vit-C & K.

cultivated varieties (North Bengal)

Cultivation method -

① ~~Cabble~~ Cabbage is winter crop, primarly grows in a temperature of 15°-20°C.
② For cabbage cultivation moisture containing heavy soils are preferred with a pH of (6-7).
③ ~~Alux~~ Nersury produced seedling are planted in cultivation ~~plant~~ land with 4-5 leaves (4-5 weeks old).
④ The spacing used in the cabbage cultivation is 40-60 cm depending on variety.
⑤ Requires consistant irrigation for moisture retention in the soil.
⑥ Requires higher amount of N for ~~to~~ leaf growth along with phosphorus & Pottasium.

For harvesting a solid and ferm heads along with few ~~leaf~~ leaves (protecting leaves) are done and marketed.

Cultivation of Okra & Cabbage major points -

① Temperature
② Soil pH
③ Seed germination
④

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Okra cultivation

① Climate & soil-

> Grows well in warm and humid climate (25°-35°c)
> Requires well-drained loamy soil rich in organic matter
> Ideal pH: 6.0-6.8 (slightly acidic)

② Land preparation-

> Plough the field 2-3 times to make the soil fine and loose.
> Add well-~~drain~~ decomposed cow dung or compost (15-20 tons/ha) before final ploughing.
> prepare ridges and furrows or raised bed for better drainage.

③ Seed selection and sowing-

> use certified, disease-free seeds (varieties: Anka Anamika, Pusa ~~sw~~ sawani)
> Seed rate → 8-10 kg/ha
> Spacing → 45cm x 30cm
> Treat seeds with ~~Toico~~ Trichoderma or fungicide before sowing.
> Sow ~~directly~~ seeds directly in the field (1-2cm depth)

④ Irrigation-

> First irrigation after sowing
> Then water every 4-5 days depending on weather
> Avoid waterlogging.

⑤ Fertilizer apply -

> Apply N:P:K = 60:40:40 kg/ha

⑥ Intercultural operation-

- weeding - After 20 days and then as needed
- Earthing up - After 30 days to support plants.
- Mulching - helps to retain moisture

⑦. Pest and disease management-

- Fruit borer - spray neem oil or insecticides
- ~~Yellow vein mosaic - Use~~

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• Yellow vein mosaic → caused by → yellow vein mosaic virus (Begomovirus) → transmitted by → white fly.

use resistant varieties like Arka Anamika and Control whiteflies.

⑧ Hervesting-

> First picking 44-45 days after sowing.
> Hervest tender fruits every 2-3 days.

Cabbage cultivation

① Climate & soil-

> Require cool climate 15-20°C
> Best in well drained fertile loam soil
> pH : 6.0 - 6.5

② Nursery preparation-

> prepare raised nursery beds (1m wide)
> mix FYM (Farm Yard Manure) and Trichoderma to prevent damping-off.
> Sow seeds thinly & cover lightly with soil.
> water regularly.

③ Transplanting-

> Seedlings ready in 25-30 days.
> Transplant in ridges
> Spacing : 45 cm x 45 cm → for early crop.
60 cm x 45 cm → for late crop.

④ Fertilizer application-

> Apply FYM → 25-30 tons/ha before transplanting
> Use N:P:K = 120:60:60 kg/ha.

⑤ Irrigation -

> water immediately after transplanting.
> maintain moisture throughout the growing period especially the head formation.

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⑥ Intercultural operations-

> weeding regularly
> Earthing up to prevent ~~tog~~ lodging & promote root growth.

⑦ Pest & ~~dist~~ disease management -

• Cabbage butterfly / diamondback moth → spray neem extract or biological control.
• Black rot → use copper fungicide if needed, avoid water stagnation.

⑧ Hervesting

> Hervest when heads are firm and compact (90-120 days after transplanting)
> cut with a knife, keeping few outer leaves for protection.

Marigold cultivation

① Climate and soil-

> Grows well in mild climate 18-30°c
> prefers well-drained loamy soil rich in organic matter.
> Ideal soil pH : 6.5 - 7.5

② Land preparation-

> Field should be ploughed 2-3 times to make the soil fine and friable
> Add well decomposed FYM (farm yard manure) 20-25 tons/ha before planting.

③ Propagation.

> marigold is propagated by seeds.
> Seeds are ~~sh~~ sown in nursery beds or trays
> Germination occurs in 4-5 days.

④ Transplanting

> Seedling are ready in 25-30 days
> Transplant at 45 x 45 cm spacing in the main field.
> Irrigate ~~imme~~ immediately after transplanting.

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⑤ Fertilizer ~~at~~ application –

> Apply N:P:K = 100:50:50 kg/ha
> Half nitrogen + full P&K as basal; remaining N after 30 days.

⑥ ~~Irrigation~~ Intercultural Operations –

> Weeding at regular intervals
> Pinching (removal of shoot tips after 30 days) encourages branching and more flower.

⑦ Irrigation –

> Water every 5-7 days, depending on weather.
> Avoid water ~~log~~ ~~lodging~~ ~~lod~~ logging.

⑧ Pest and disease Control –

> Aphids & mites → ~~th~~ spray neem oil or insecticide.
> Leaf spot → use copper fungicide.

⑨ Hervesting –

> Flowers are ready to harvest in 60-70 days after transplanting.
> Pick fully opened flowers in the morning for best freshness.

Tuberose cultivation

① Climate and soil –

> Thrives in worm, humid climate (20°-30°c)
> Requires ~~wea~~ well drained sandy loam soil rich in organic matter.
> pH between – ~~6.5~~ 6.5-7.5 is ideal.

② Propagation –

> Propagated by bulbs, not seeds.
> Select healthy, disease free bulbs (2-2.5 cm diameter)

③ Land preparation –

> Plough the land 2-3 time, make it fine and level.
> Mix FYM 20-25 tons/ha before planting.

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④ Planting –

> Planting time – February-March (Spring) or
July – August (rainy season)

> Spacing – 30x20 cm

> Bulbs are planted 5-6cm deep with tips upword.

⑤ Fertilizer application –

> Apply N:P:K = 200:100:100 kg/ha

> Half Nitrogen + full P&K at planting, rest N after 45 days.

⑥ Irrigation –

> water immediately after planting

> Then irrigate every 7-10 days regularly.

> Avoid water-logging.

⑦ Intercultural operations –

> weeding & hoeing regularly.

> Earthing up after 40 days to support plants & bulbs.

⑧ Pest and disease control –

> Thrips & aphids – spray neem oil or dimethoate

> Bulb rot → treat bulb ~~before~~ with fungicide (carbendazim) before planting.

⑨ Harvesting –

> Flower spikes after 3 months of planting

> Harvest when 2-3 flowers on the spike open.

> Use for garlands, Bouquets & perfume industry.

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		Okra	Cabbage	Marigold	Tuberose
① C & Soil	Temp	25°-30°C	15°-20°C	18°-30°C	20°-30°C
	Weather	warm & Humid	cool	mid	warm & humid
	pH	6.0-6.8	6.0-6.5	6.5-7.5	6.5-7.5
② Fertilizer apply	NPK ratio	60:40:40 kg/ha	120:60:60 kg/ha	100:50:50 kg/ha	200:100:100 kg/ha
③ Pest & disease		• Fruit borer • YVMV	• Cabbage butterfly • diamondback moth • Black rot.	• Aphids & mites • Leaf spot	• Thrips & Aphids • Bulb rot
④ Hervesting period -		45 days after sowing	90-120 days after transplanting	60-70 days after transplanting	3 month after planting

Okra
① Climate & soil
② Land preparation
③ Seed selection & sowing
④ Fertilizer apply
⑤ Irrigation
⑥ Intercultural operation
⑦ Pest & disease control
⑧ Hervesting

Cabbage
① Climate & soil
② Nursery preparation
③ trans planting
④ Fertilizer apply
⑤ Irrigation
⑥ Intercultural operation
⑦ Pest & disease control
⑧ Hervesting

Marigold
① Climate & soil
② Land preparation
③ Propagation
④ Transplanting
⑤ Fertilizer apply
⑥ Irrigation
⑦ Intercultural operation
⑧ Pest & disease control
⑨ Hervesting

Tuberose
① Climate & soil
② propagation
③ Land preparation
④ Planting
⑤ Fertilizer apply
⑥ Irrigation
⑦ Intercultural operation
⑧ Pest & disease control
⑨ Hervesting

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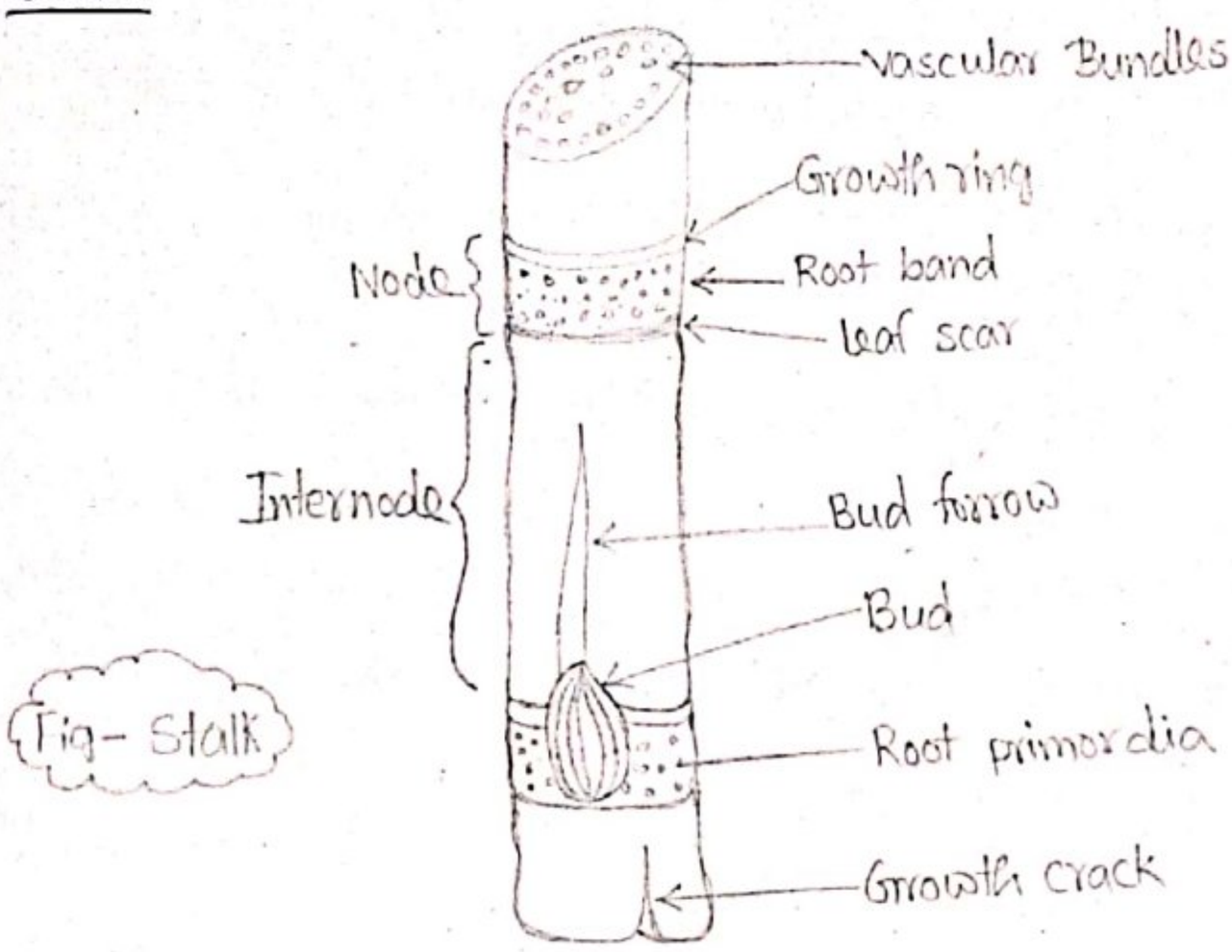
sugercane

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Morphological characters of sugercane :-

Stalk :



① The stalk consists of segments called joints. Each joint is made up of a node and an internode.

② The node is where the leaf attaches to the stalk and where the buds and root primordia are found.

③ The length and diametre of the joints vary widely with different varieties and growing conditions.

④ One bud is present on each node, and they alternate between one side of the stalk to the other.

⑤ Variations in size, shape and other characteristics of the bud provide a means of distinguishing between varieties.

⑥ When seed-cane is planted, each bud may from a primary shoot. From this shoot, secondary shoots called "tillers" may from the underground buds on the primary shoot.

⑦ In turn, additional tillers may form from the underground secondary shoot buds.

⑧ The colours of the stalk at the internodes depend on the cane variety and environmental conditions.

⑨ Exposure of the internodes to the sun may be ~~rest~~ resultant in a complete change of colour.

⑩ All colours of the stalk derive from two basic pigments -
- The red colour of anthocyanin.
- The green " " chlorophyll.

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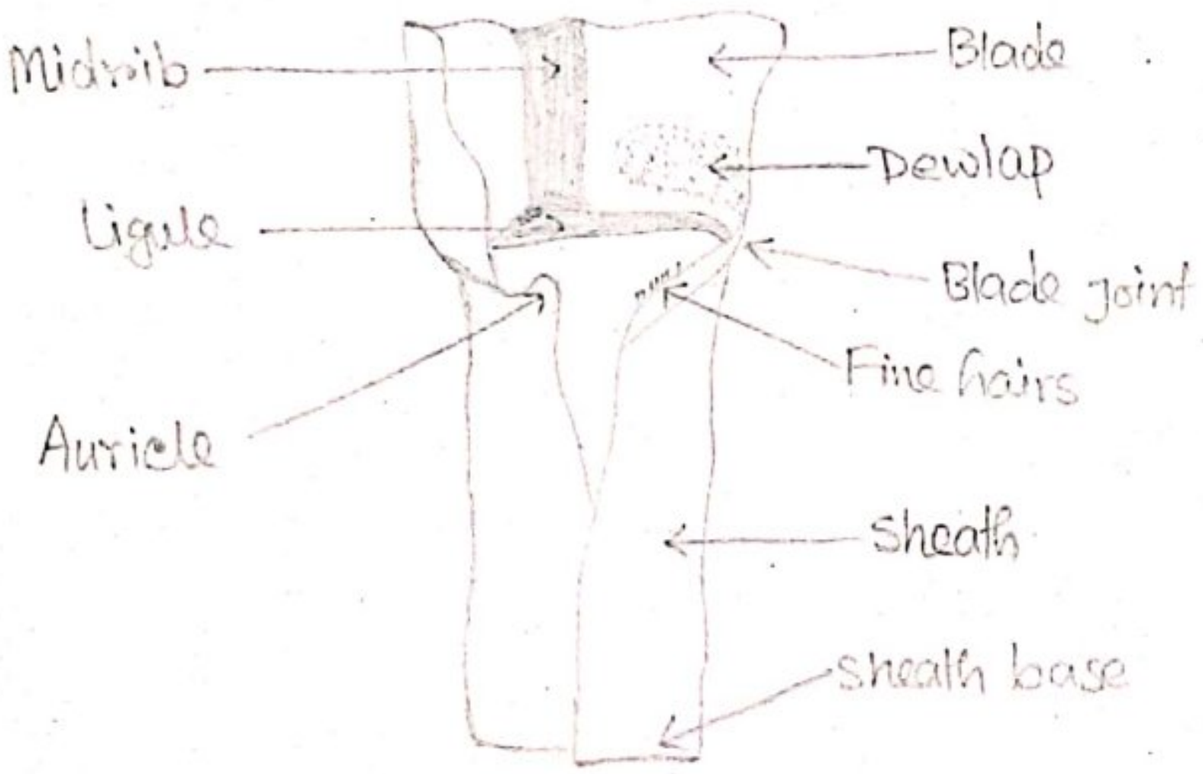
- The ratio of the two pigments produce colours from green to purple-red to red to almost black.

⑪ Yellow stalks indicate a relative lack of these pigments.

⑫ The top of the stalk is relatively low in sucrose but 1/3 contains many buds and a good supply of nutrients which makes it ~~valu~~ valueable as seed cane for planting.

⑬ Two types of cracks are sometimes found on the surface of the stalk-
> harmless, small ~~cra~~ cracks restricted to the epidermis.
> harmful, deep crack, run the whole length of the internode.

Leaf :



The leaf of sugercane plant is devided into two parts
↓
~~Seath~~ Sheath & Blade → seperated by blade joint

① leaves are usually attached alternately to the nodes.
② Mature sugercane plant has an average total upper leaf surface of about $(0.5)^2 m$
③ The number of green leaf / stalk is about 10, depending on variety and growing condition.

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Systematically represent the process of commercial production of cane suger.

»

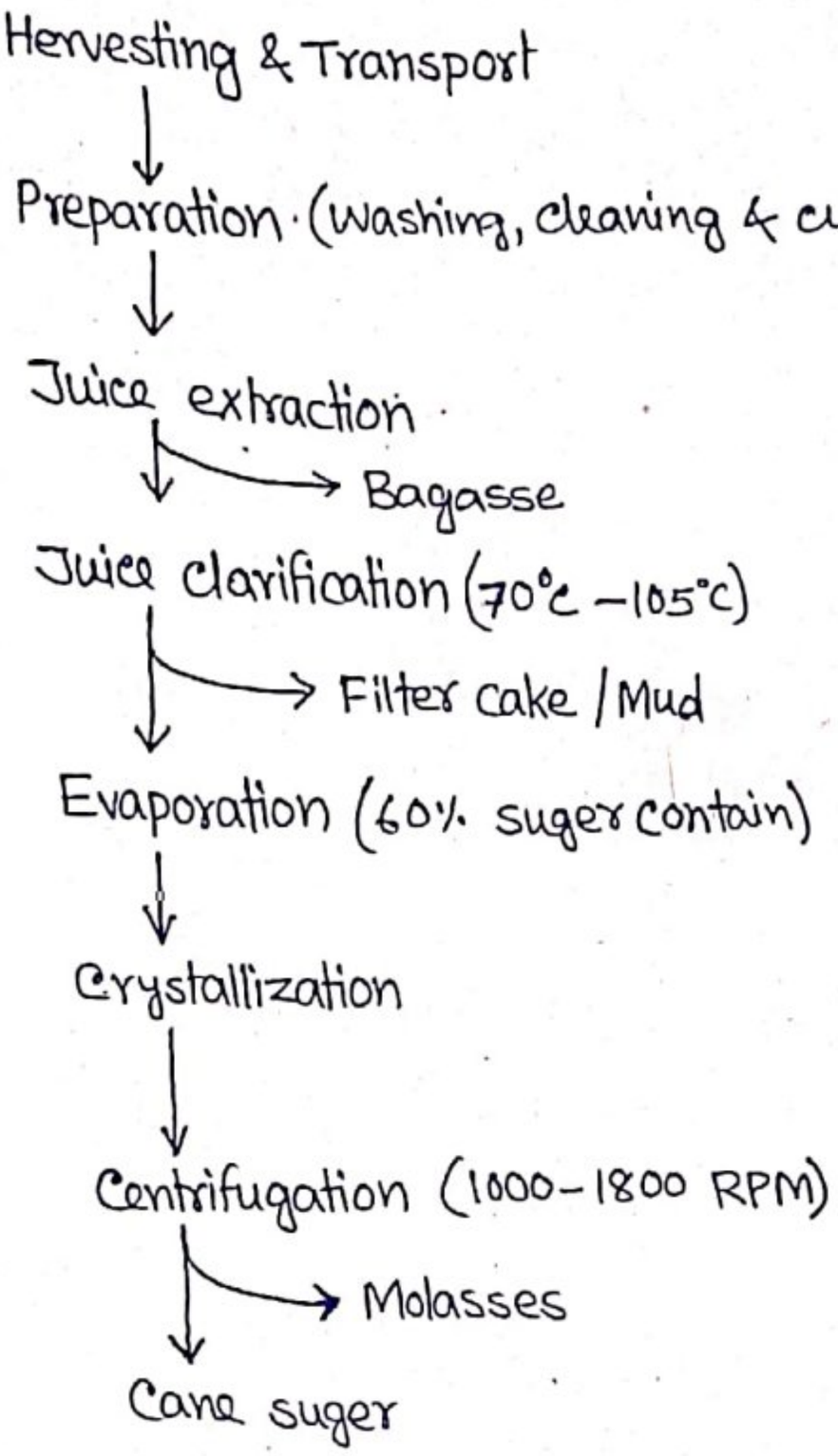


Fig- Systematic representation of Sugercane processing.

① Hervesting & Transport

Sugercane is cut when it is mature (usually after 10-20 months). Hervesting can be manual (by hand) or mechanical (by machine). Then transport to suger mill/factory.

② Washing & preparation

The hervested canes are washed to remove mud, leaves and dirt. They are cut into small pieces to prepare for juice extraction.

③ Juice extraction

The prepared cane is crushed in rollers or mills to extract the juice. The cane is passed through 4 rollers and to get maximum juice, hot water is sprayed ~~in the~~ when passing through 4th roller.

The leftover fibrous material is called baggasse.

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4. Juice Clarification

The extracted juice contains impurities. Extracted juice mixed with water is sent to boiling house and juice is heated in juice heater at 70°C. In order to bleach and maintaining ph of the juice it is treated with the milk of lime & Sulphur dioxide. The juice is further heated at 105°C, then the juice goes into purifier tank. It takes over two hour to settle impurities ~~to the~~ at the bottom of the tank.

The settle mud is pumped to rotry drum vaccum filter where the mud is filtered from the juice & filter juice is return to process further. while the press mud cake remains outside.

5. Evaporation

The clarified juice is concentrated by evaporating water in multiple effect evaporators. The syrup obtained contains about 60% suger.

6. Crystallization

The concentrated syrup is colled and seeded with fine sugar crystals to form sugar crystals. This mixture of crystals and molasses is called massecuite.

7. Centrifugation

The massecuite is spun in (high RPM between 1000-1800) centrifugal machine to seperate suger crystals from molasses.

8) Drying & Grading

The seperated suger crystals are dryed using hot air and graded ~~or~~ according to particle size.

9) Packaging and Marketing

The final cane suger is ~~pag~~ ~~packed~~ weighed, and packed for marketing.

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By products of sugercane Processing

By Products	Use
① Bagasse ↓ (remaining fibrous material after juice extraction)	Used as fuel in boilers, paper and ~~fibre~~ fiberboard industries.
② Molasses ↓ (remaining liquid material after centrifugation)	Used in alcohol & ethanol industries, Animal feed.
③ Filter cake (press mud) ↓ (remaining solid impurities after juice purification).	Used as fertilizer

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Tea & Coffee

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Tea

Common name- Tea (Cha)
Scientific name- Camellia sinensis
Family name- Theaceae.
Parts used- Two tender leaves and an unopen bud.

native country- China

Major cultivated varieties

① Camellia sinensis var. sinensis → Chinese tea.
② Camellia sinensis var. assamica → Assam tea.

Assam tea	china tea
① It is a tree.	① It is a shrub.
② Large glossy leaves.	② Small leathery leaves.
③ Light- medium green colour.	③ Dark green in colour.
④ ~~Hel~~ High yeild & medium quality.	④ low yeild & good quality.

Morphology of Tea :-

Tea is a plant of evergreen, woody and shrub category. Usually maintained at about 1m height for commercial plantation & but in wild it can grow about 15 m height.

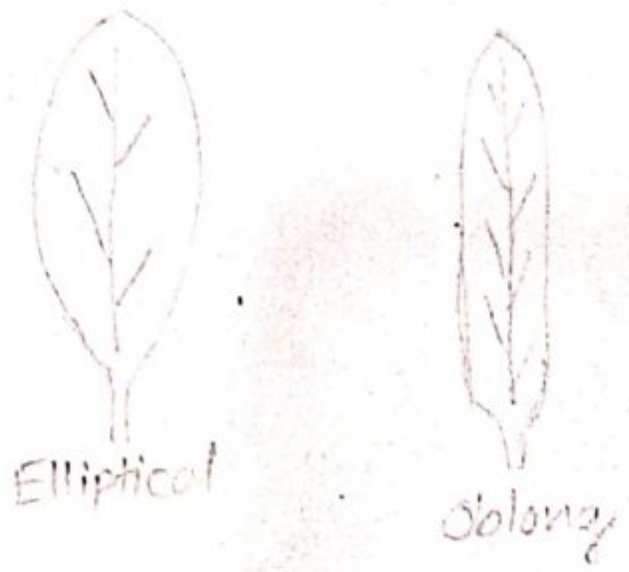
Leaves

Tea leaves are commercially use for the reproduction of young leaves and terminal buds). Shape of leaves are typically elliptical, Oblong or elliptical oblong with leathery texture & short petioled.

Size of the leaf ranging b/w 4-22cm in length. leaf margin are serrated to serrulated.

The upper surface of the leaf is hairless & dark green in colour. The lower side of the leaf are pubescent (covered with fine hairs).

leaf venation comprised of central vein raise on both side & smaller 7-9 distinct lateral veins covering the leaf blade.



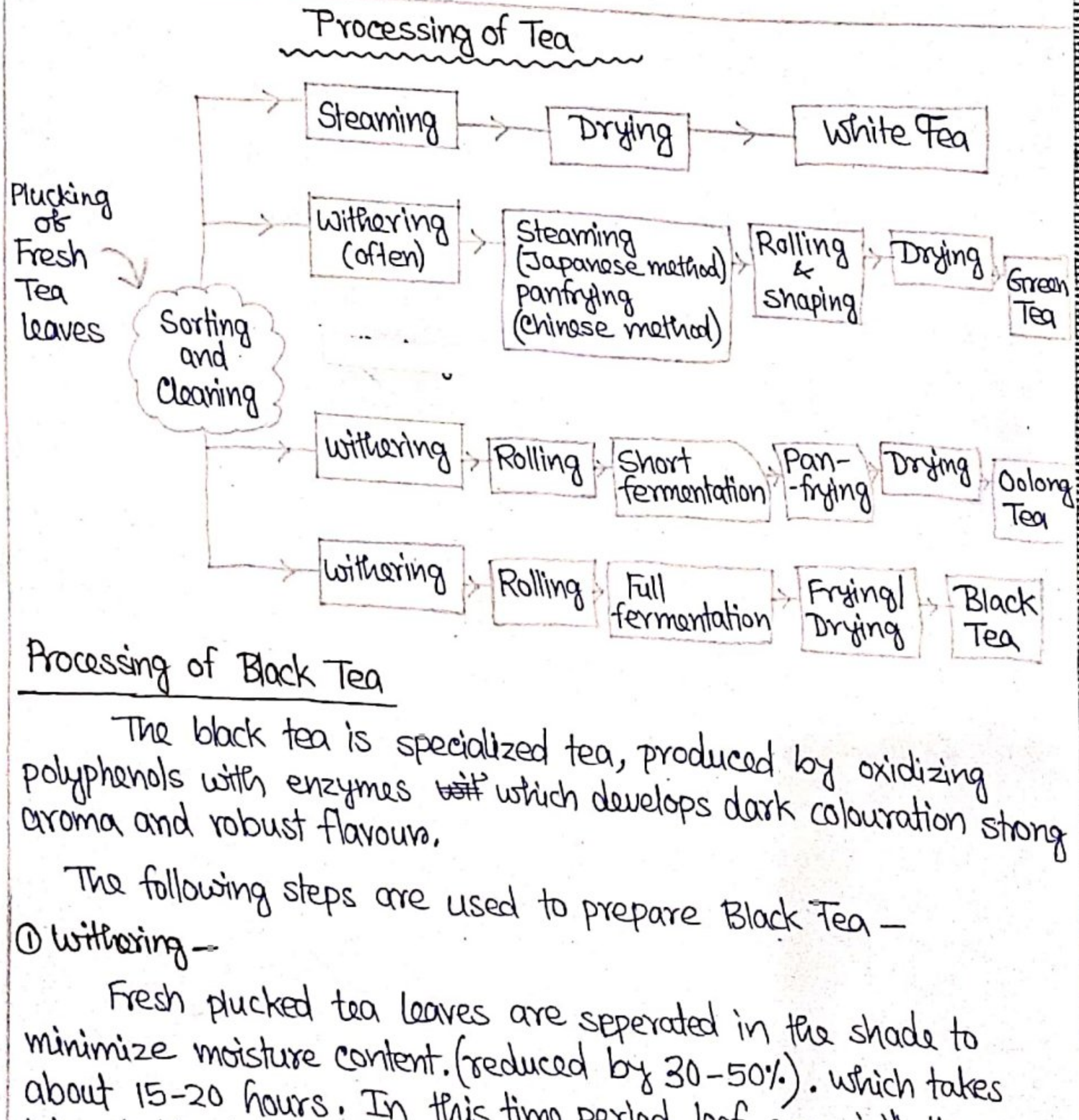
Give a whole tree figure.

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Flowers :-

Tea plants produces white fragnant flowers length about 2-4cm with 6-8 petals & grows in the leaf ~~axiles~~. axils.



Processing of Black Tea

The black tea is specialized tea, produced by oxidizing polyphenols with enzymes ~~usit~~ which develops dark colouration strong aroma and robust flavour.

The following steps are used to prepare Black Tea –

① Withering –

Fresh plucked tea leaves are seperated in the shade to minimize moisture content. (reduced by 30-50%). which takes about 15-20 hours. In this time period leaf saps initiating internal biochemical changes.

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② Rolling or ~~distus~~ distruption —

The welted tea leaves are rolled manually or mechanically by ~~following~~ CTC (crush, Tear, Curl) method, to physically ~~break~~ break the cell walls of leaf and release the internal enzymes and ~~essencial~~ essential oils. This internal enzymes upon expose to oxygen iniciate the ~~polyphenols~~ oxidizing ~~ph~~ polyphenols of the tea leaf.

③ Oxidation / Fermentation-

The old tea leaves spread into the climate controll room (with high humidity and temperature) for several hours. The mid enzyme polyphenol oxidase reacts with leaf polyphenols causing the leaves turn into brown colour or copperate colour.

④ Drying / Frying -

The ~~oxidised~~ oxidized ~~leafs~~ ~~passe~~ leaves passed through hot air (80-90°C) to holt the oxidation process instantly and reduce the moisture content at the level of 3-5% which locks the tea aroma and flavour.

⑤ Packaging ~~and~~ and marketing

The final CTC leaves then packaged and ~~marks~~ marketed for selling.

Processing of Green Tea

The green tea is another popular category of tea which is non fermented or unoxidized, retaining green colour fresh flavoury and higher leaves of naturally occuring polyphenols, amino acid L-theanine various vitamins and minerals.

The following steps are used to prepare Green Tea-

① Fixation -

After plucking fresh tea leaves quickly heated to deactivate polyphenol oxidase enzyme and blocks the oxidation / fermentation.

- Tea leaves are steamed → in Japanese method.
- Tea leaves are Panfried / Rosted → in chinese method.

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① Add 'sorting & Grading' before 'packging & marketing'— Based on the leaf colour, aroma the tea sorted and graded.

② Rolling —

~~The~~ ~~fixed~~ ~~tea~~ ~~leaves~~ ~~rolled~~ ~~and~~ ~~shaped~~ ~~niddle~~, ~~pellets~~

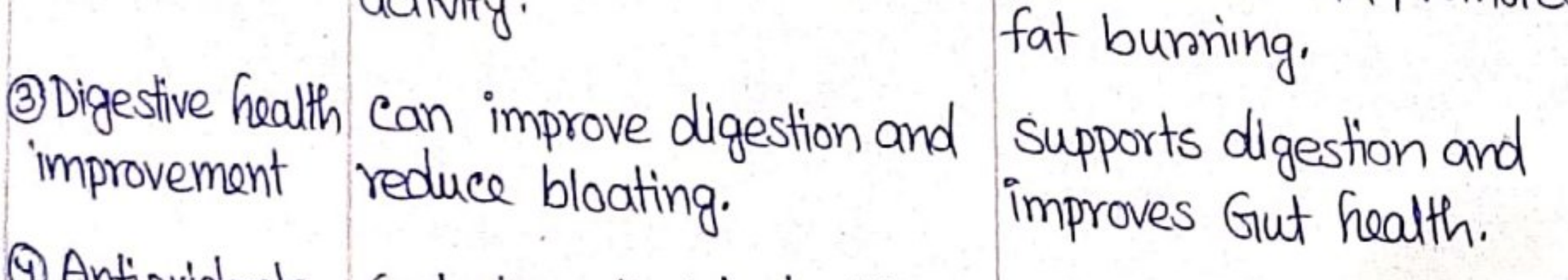
The rolling process of green tea, done after fixing, shapes the leaves (like needles or pellets) by twisting and breaking cell walls gently and release essential oil to intensify the tea flavour.

③ Drying-

The rolled leaves then dryed to remove moisture and preserve the aroma and flavour of the tea leaves.

④ Packaging and marketing —

Then, The dryed leaves are packaged and marketed.



Uses	Black Tea	Green Tea
① Primary use	Highly popular daily beverages, hot often consumed ~~with~~ with milk and sugers.	Popularly used as ~~daily~~ daily beverage. (without sugers).
② Weight management	Contribution for hydration and supports overall metabolic activity.	Often link with boost of metabolism and promotes fat burning.
③ Digestive health improvement	Can improve digestion and reduce bloating.	Supports digestion and improves Gut health.
④ Antioxidents	Contents antioxidents like polyphenols, theaflavins, thearubigins etc.	very rich in antioxidents like catechins.
⑤ Improves mental alartness	rich in caffine and ~~Thiorine~~ theanine improves mental alartness (focus and clearity of thoughts).	Also contains high amount of caffine and theanine.

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COFFEE

Common name- Coffee
Scientific name- Coffea arabica.
Family name- Rubiaceae
Parts used- Seed (Beans)
native country- Ethiopia. (Historically Abyssinia) (Today)

Out of several species, two prominent categories of Coffee observe in global market.

① Coffea arabica → Arabian coffee.
② Coffea robusta → Congo coffee.

Morphology-

Plant habit-

It is a tropical evergreen, small tree or shrub plant, height can be reached upto 10 m tall, (commertially 2-3m).

leaves-

Simple, opposite and elliptical in shape, appears dark green in colour, glossy surface with waxy pointed tip. (length varies in range between 5-20 cm).

Flowers-

Produces small white highly fragnant flowers & grows in axillary clusters at the leaf nodes.

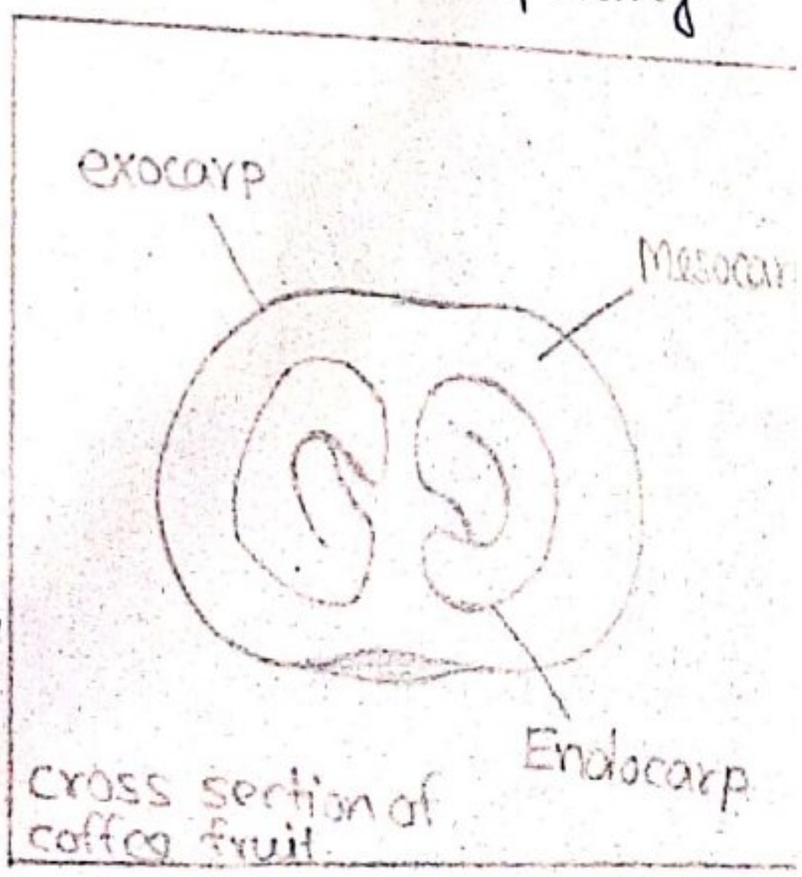
Fruits

produces drupe type of fruit, initially green in colour and eventually turns into bright yellow in colour and after ripening turns into crimson red.

exocarp-(outer skin) - Green when immature, typically turns bright yellow or red when ripen.

Mesocarp - located beneath the exocarp, middle layer, when ripe it become sticky, pulpy, gelatenous layer clo mucilage.

Endocarp - This is innermost layer, protective shell that sorrounds the coffee seed.



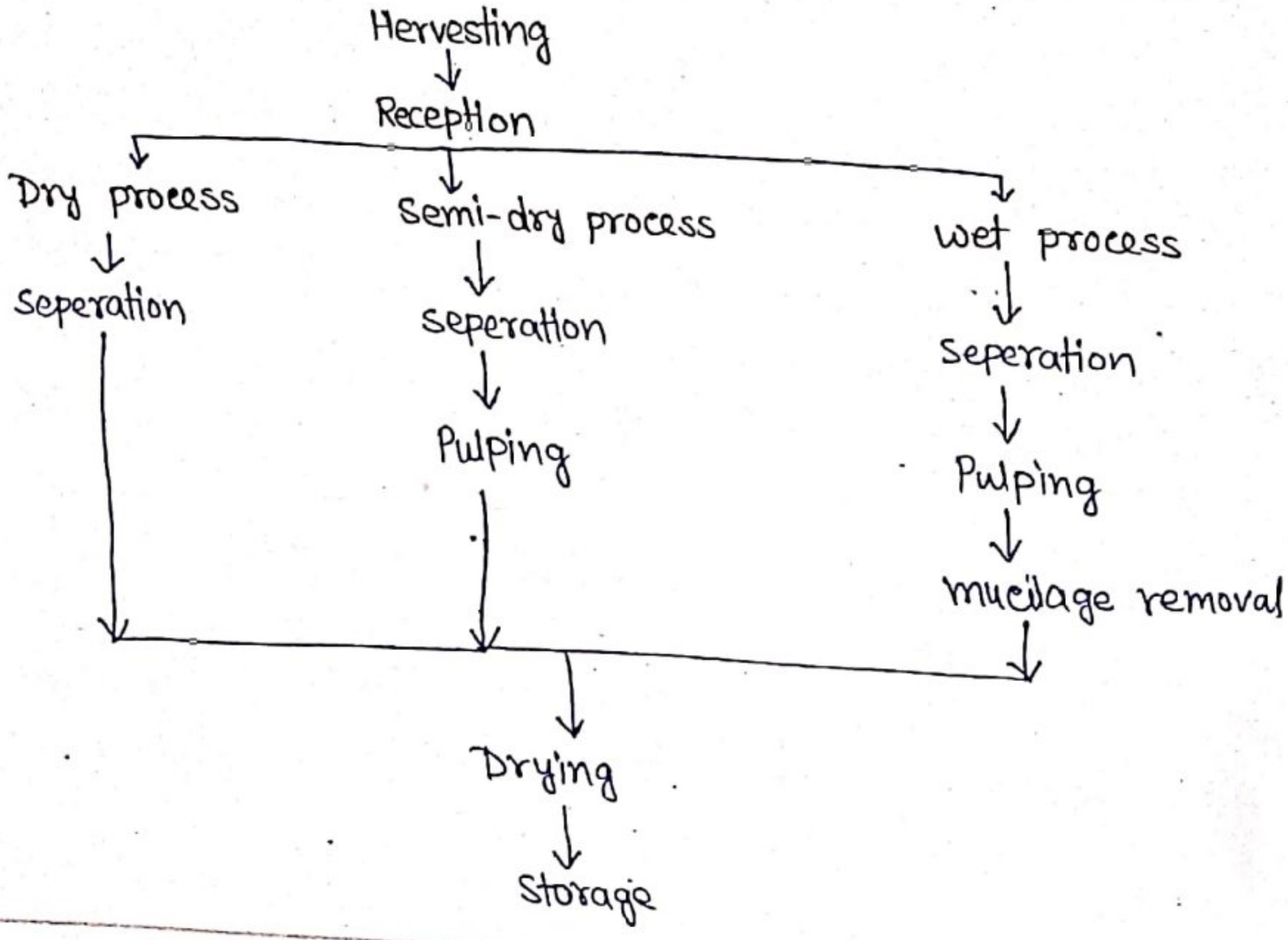
cross section of coffee fruit

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Seed :-

A single coffee fruit (coffee cherry) contains two seed, facing each other, coted with thin layer of spermoderm (thin membrane)

Processing of Coffee.



Processing of coffee

- Dry process
  - > Dry whole cherry.
  - > full fruity body.
  - Fruity flavour
- Semi dry process
  - > Dry after removing of fruit skin
  - > But left some mucilage
  - Honey flavour
- wet process
  - > Dry after removing of fruit skin and also remove the mucilage covering
  - floral flavour

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Processing of Coffee-

① Dry process-

① The whole seeds spred in thin layers under sun for 3-6 weeks to prevent mould attack.

② The beans absorb suger from the drying fruit leading to intense flavour and low acidic.

② Wet process-

After removing beans from the fruit the seeds fermented in water tanks for 12-72 hours to break the mucilage covering and washed properly before drying result in high acidic and distinct floral flavour in from those coffee beans.

③ Semi dryed / Pulped processing

In this process, the skin of the fruit is removed but not mucilage covering removed entirely (left some of the mucilage called honey). Then sundryed and prepare a natural coffee

Post processing / Rosting:-

Green beans heated at 180° - ~~210~~ 240° c for triggering chemical reaction (millad rxn) which develops a distinct aroma and brown colouration of coffee beans.

Grinding:-

The rosted beans are grinds into various sizes.

① Coarse
↓
> large particle
> primarily used in the ~~coff~~ cold coffee.

② Medium
↓
> widely used for dripped coffee

③ fine particles
↓
> Used for the preparation of ~~esp~~ espresso.

Millard Rxn - is a vital complex chemical process in cooking where amino acids (from protein) & reducing suger react with heat, creating complex flavour, apitizing brown colour and rich aromas in food like coffee, ~~bea~~ baked bread and toasted ~~mashroom~~ mash mallows

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Drug & narcotics

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Plant	Scientific name	Family	Main drug	Use	Health Hazards
① Cinchona	Cinchona officinalis	Rubiaceae	Quinine	Anti-malarial	~~Nausia~~ Nausea, blurred vision.
② Digitalis / Fox glove	Digitalis purpurea	Scrophulariaceae	Digitalin, Digitoxin, Digoxin (cardiac glycosides)	Heart treatment	Overdose causes Vomating, irregular heart beat.
③ Papaver / Opium poppy	Papaver somniferum	Papaveraceae	Morphine, codeine, Thebaine	Painkiller, Sedative, cough relief, anesthesia	Addiction, Constipation, mental depression
④ Cannabis / Ganja (Indian Hemp)	Cannabis sativa	Cannabaceae	THC (Tetrahydro-cannabinol) $C_{21}H_{30}O_2$	Sedative, ~~analgesic~~ analgesic	Addiction Hallucination memory loss
⑤ Tobacco	Nicotiana tabacum	Solanaceae	Nicotine	Stimulant, limited medical use.	Addiction Lung cancer heart disease.

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Groundnut

coconut

soybean

mustard

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Groundnut → Arachis hypogeal → Fabaceae (leguminosae)

edible part → seed / kernal

uses –

① Edible oil production
② Food products → sweets, snacks
③ Animal feed → groundnut cake
④ Soil improvement → fix nitrogen

Coconut → Cocos nucifera → ~~Area~~ Arecaceae (Palmae)

edible part → endosperm

uses –

① Edible oil production
② Food products → sweets,
③ Cosmetics & medicinal use → coconut oil → use for making soaps, hair oil & skincare products
④ Household & industrial prod → mat, roop etc.

Soybean → Glycine max → Fabaceae (leguminosae)

edible part → Soyabean seeds

uses –

① Edible oil production
② food products → soy milk, tofu, soy sauce, soy chunks
③ Animal feed → Soybean meal (after oil extraction) is a major ingredient in poultry & livestock feed.
④ Industrial use → Cosmetics, paints.

Mustard → Brassica campestris → Brassicaceae

edible part → leaves & seeds

uses –

① Edible oil production
② Medicinal uses → traditionally use for joint pain
③ Green manure & fodder → mustard plant can be used as green manure to enrich soil & as fodder for cattles.
④ Spices → use for flavouring agent

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Sal & Segun

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ϕ Common answer for Sal & Segun. (Replace Sal/Segun Acc to Question)

> Sal (Shorea rubusta) / Segun (Tectona grandis) is one of the most important forest trees of West Bengal, ~~particallya~~ found in ~~the~~ mainly Bankura, Purulia, Jalpaiguri, Coochbehar, Birbhum districts.

General Account :

- Sal forests are tropical moist and dry deciduous in nature.
- Trees grows well in well-drained, slightly acidic sandy loam soils.
- It is a hard wood tree
- It can grow up to 30-35m (for sal) / 40-45 m(for segun)

Economic Importance-

① Timber – Sal wood is extremely hard, durable & termite resistant, so widely used for furniture, constructions etc. (doors, windows)

② Revenue – Sal plantation provide significant economic ~~resu~~ return to the state & local fermers.

③ Employment – Provide livelihood opportunities in logging and timber industries.

Ecological Importance –

① Soil conservation – Dense root system helps prevent soil erosion & maintain soil fertility.

② Carbon sink – Sal forests act as significant carbon reservoirs, reducing atmospheric CO_2

③ Biodiversity – Supports a rich variety of flora & ~~funa~~ fauna, maintaining ecological balance.

④ Micro climate control – Provide shade and moisture retention improving local environmental conditions.

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some remaining topics

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① NBPGR – National Bureau of Plant Genetic Resources. (HQ - New Delhi)
② FAO – Food and Agriculture Organisation. (HQ - Rome, Italy)
③ CGIAR – Consultative Group on International Agricultural Research (HQ - Montpellier, France)

Q) What is Plant introduction?
→ TPS → True Potato Seed.

④ ICAR → Indian Council of Agricultural Research.
⑤ BSI → Botanical Survey of India.
⑥ FRI → Forest Research Institutes.

Q) Four timber yeilding plant of NB.
> ① Sal → Shorea robusta → Dipterocarpaceae
② Teak → Tectona grandis → Lamiaceae
③ Gamari → Gmelina arborea → lamiaceae
④ Shimul → Bombax ceiba → Malvaceae
⑤ Mahogany → Swieteria mahagoni → Maliaceae × Meliaceae

Q) Four citrus family/species other than mandarin orange –
① Lemon → Citrus limon →
② Lime → Citrus aurantiifolia →
③ Grapefruit → Citrus paradisi →
④ Sweet orange → Citrus sinensis →
} family Rutaceae (citrus family)

Q) What is ethenomedicine?
> Based on indigenous knowledge, study of plants to cure or treat disease by ethenic people is called ethenomedicine

As per WHO (46 words) (8 commas)
"The sum total of the knowledge, skill, and practices based on the theories, beliefs, and experiences indigenous to different cultures, wheather explicable or not, used in the ~~maintain~~ maintenance of health as well as in the prevention, diagnosis, improvement or treatment of physical and mental illness."

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3) Name the enzyme use for oxidation/fermentation of Black tea?

> Enzyme - Polyphenol oxidase (PPO)

Tea Polyphenol (catechins) ⟶ theaflavins & thearubigins.

> During the fermentation process of black tea Polyphenol oxidase catalyzes the oxidation of catechins (tea polyphenol) into theaflavins & thearubigins

> These compounds ~~gave~~ give black tea its colour, flavour and aroma.

---

9) Why rolling is necessary during processing of Tea?

> Rolling is necessary for two main reasons-

① To rupture cell wall — It breaks the cell wall of leaf and release the juice/saps, which contain the ~~tea~~ enzymes and polyphenols needed for flavour

② To initiate oxidation in case of black tea

By exposing these juice to the air, rolling starts the oxidation (fermentation) process that ~~def~~ develops the flavour and aroma.

---

3) Why withering is necessary during processing of Tea?

> In tea processing, withering is the first essential step where freshly plucked leaves are spread out to reduce moisture content upto 30-50%.

① Reduction of moisture content.

② Development of Aroma & flavour.

③ Activation of enzyme PPO

④ Reduction of leaf brittleness → make leaf Soft & flexible, suitable for rolling.

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9) What is ethnoecology?

Ethnoecology → Ethno + Ecology
↓ → relationship b/w living thing & their environment
People/culture

> Ethnoecology ~~means~~ is the study of how different human cultures understand, use & manage their natural environment — including plants ~~and~~, animals and soil, ecosystems.

ex →

① Tribal communities knowing which plants are medicinal.

② Fermers use local knowledge for crop rotation or pest control.

③ Indigenous people identifying animal behaviour to predict ~~the~~ weather.

It helps to ~~protect~~ & preserve Traditional Ecological Knowledge (TEK)

① Rice → aromatic → Gobindobhog, Kataribhog.
② Rice → IR-36, IR-64, Swarna
③ Wheat → HD-3118, HD-2733, Sonalika, kalyansona.
④ Sugercane → Co-775, Co-740
⑤ Potato → Kufri Jyoti, Kufri chandramukhi, Kufri surya.
⑥ Tea → China tea (CSVS), Assam tea (CSVA)
⑦ Coffee → Coorg Arabica (Karnataka), Araku valley Arabica Coffee (Andhra pradesh)
⑧ Tobacco → GCT-2, Sona
⑨ Cinchona → C-officinalis, C ledgeriana
⑩ Jute → JRO-524, JRO-7835, JRC-321
⑪ Okra → Arka Anamika, Pusa sawani, prabhani kranti
⑫ Cabbage → Golden Acre, pusa mukta, pusa drum head.
⑬ Marigold → pusa Narangi, Gainda, pusa basanti Gainda
Siracole Orange

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⑭ Tuberose → Prajwal, shrinagar, suvasini.

⑮ Mango → Himsagar, Fazli, Langra, Dashehari., Chausa, Malika

⑯ Orange → Darjeeling mandarin,
Khasi "
Coorg "
Nagpur orange

Q) what do you mean by Ratoon Crop?

> A ratoon crop is the crop that grows from the stubble (remaining part of stem) of the previous ~~harvesting~~ harvested crop without replanting new seeds or seedlings. It is also c/o regrowth crop or second crop.

benifits

> The practice saves time, labour and cost of cultivation

example

sugercane, banana, p'neapple are common ratoon crop.

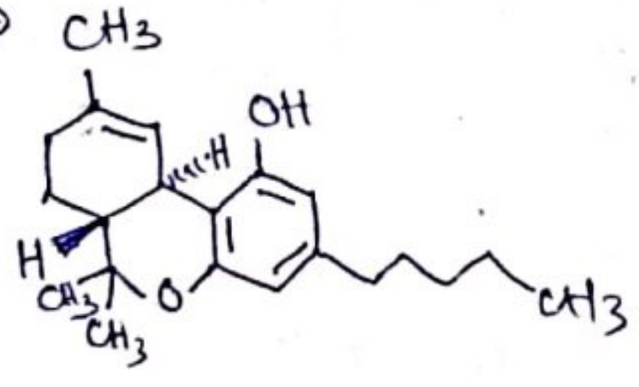
Q)

Cropping Season	Sowing Time	Harvesting Time	Yeild/Acre	Irrigation
Autumn (Aus)	May–June	September–October	≈ 500 kg	lightly irrigated
Winter (Aman)	June–July	November–December	≈ 870 kg	Depends on Rain, no irrigation re
Summer (Boro)	November–December	April–May	≈ 1300 kg	Depends fully on irrigation.

Q) Phytochemical in Ganja –

THC → Tetra hydrocannabinol (Δ9 THC)

Structure →



~~C21 H30 O2~~

$C_{21} H_{30} O_2$

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① Define ethnomedicine & as per WHO.
② what is ethnoecology?
③ Define Domestication.
④ Define Genetic ~~erot~~ eroslon.
⑤ Define Germplasm.
⑥ What is plant introduction?
⑦ What ~~is~~ ~~is~~ are the types of plant introduction?
⑧ What are the objectives of plant introduction?
⑨ procedure of plant introduction?
⑩ what is acclinatization?
⑪ What ~~is~~ are cereals? 2 example
⑫ What are millets? 2 example
⑬ What are legumes? 2example
⑭ What are fodder legumes? 2example
⑮ What do you mean by Ratoon crop?
⑯ Four Timber yeilding plant of N.B. (with sci name).
⑰ Four citrus family other than mandarin orange.
⑱ why rolling is necessary during Tea processing?
⑲ " withering " " " " " " ?
⑳ Name the ~~varie~~ enzymes use for oxidation of Black tea?
㉑ Aus rice ? time ?
㉒ Aman rice ? time ?
㉓ Boro rice ? time ?

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① Rice → Swarna, IR-36, ~~Katak~~ Kataribhog, Gobindobhog
② Wheat → ~~Solan~~ Sonalika, Kalyansona
③ Jute → JRO 632, JRC 517
④ Okra → Pusa A-4, Kashi kranti, kashi lalima, Pusa Makhmal
⑤ Cabbage → Golden Acre, Pusa Drum Head, Pride of India
⑥ Mango → Fazli, Dasherhari, Langra, Himsagar
⑦ Orange → Darjeeling Mandarin, Nagpur orange.
⑧ Tea → Darjeeling tea, Assam tea
⑨ Coffee → Coorg Arabica coffee → Karnataka.
↳ Araku valley Arabica coffee → Kerala.

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Q) What is Cereals ?

> Cereals are grasses cultivated for their ~~edible~~ edible, starchy grains, serving as major ~~staple~~ staple foods worldwide. ~~including wheat~~

ex → wheat, (Triticum aestivum)
Rice (Oryza sativa)

Characteristics of Cereals

① The are mainly poaceae family / Grass family plants previously known as Graminae.
② They provide a significant portion of the world's food, majorly known as staple food.
③ They are rich in carbohydrates, fiber, protein & minerals

Uses

① primarly use as whole grains in case of rice.
② Wheat flour used for bread, pasta etc.
③ Major breakfast of almost all over the world.
④ Use to get carbohydrates for daily work.

Q) What are legumes ? (High protein containing plant family)

> legumes are plants in the pea/bean family (Fabaceae) grown for their edible pods or seeds.

ex → chick pea (cicer arietinum), Pigeon pea (Cajanus cajan)

Q) What are fodder legumes ?

> Fodder → Any crop grown for livestock feed.

legumes → Plants like alfalfa, clover, that produce pods & are rich in protein and fibre.

Fodder legumes → Growing plants from the legume family especialy to feed livestock for nutrition & their high protein. They have ability to improve soil by fixing ~~atos~~ atmospheric N_2 Nitrogen in their roots.

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Q) What are millets ?

> Millets are a diverse group of small-seeded plants, placed in grass family (Poaceae). They are cultivated for thousands ~~of~~ of years for being highly nutritious.

Common types

	Scientific name
① Sorghum (Jowar) →	Sorghum vulgare
② Pearl millet (Bajra) →	Pennisetum typhoides
③ Finger millet (Ragi) →	Eleusine coracana

Importance of millets -

① They are eco-friendly, require less water than rice or wheat.

② They ~~have~~ are ~~not~~ health benificial, especially for diabetis and anemia

③ They have global significance, over half a billion people use for their primary food.

④ Organisations like FAO have taken initiative for their role in food security.

Q) Define Domestication.

> Domestication is the process of adapting wild plants and animal for human use.

Domestication is the evolutionary, multi-generational process of bringing wild plant species under human control, altering them genetically and morphologically to meet human needs.

Aspects of plant domestication

① Genetic & phenotypic Alteration - Domesticated plants differ significantly from their wild ancestors due to human-driven rather than natural selection.

② Human Dependency - Domesticated many species loss their natural dispersal mechanism and require cultivation and propagation.

③ Process - It is a gradual, intentional process b/w human & plants.

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g) Queen of spice → mom → Cardamom
g) king of spice → father → spicy → Black paper.

g) What is genetic erosion?

> Genetic erosion is the gradual loss of genetic variation in a species, especially among cultivated plants and domesticated animals.

Cause of Genetic erosion

① Replacement of traditional varieties with modern high-yield varieties.
② Distruction of natural habitats — deforestation, urbanizatio
③ Over ~~expel~~ exploitation of contain species.
④ Climate change and environmental stress
⑤ Pollution and use of chemicals.

ex → Farmers stop growing local rice or wheat varieties and switch to hybrid ones — the old genetic varieties disappea
→ wild relatives of crops get lost due to habitat distruction.

g) Define Germplasm:

> Germplasm is the total collection of genes or genetic material available in a species, which can be used to improve future generations.

Germplasm refers to the living genetic resources such as seeds, tissues, or other plant or animal materials that are used for breeding, conservation and research.

ex → Seed stored in gene banks, tissue cultures ~~used~~ } → used to preserve crop varieties.

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Essential oil V/s Fatty oil

	Essential oil	Fatty oil
1) Source	1) Distilled from plant part.	1) Processed from seeds.
2) Molecular size	2) Tiny molecule	2) large molecule
3) Aroma	3) Aromatic & volatile	3) Non aromatic & nonvolatile.
4) Circulation through plants & human bodies	4) Circulate throughout plants & human bodies.	5) Donot circulate throughout plants & human bodies.
5) Circulation through tissue & cell wall.	5) Can pass through tissues, cell walls & cell membranes.	5) Cannot pass through tissues cell walls and cell membranes.
6) Nature	6) Not greasy to the touch	6) Greasy to the touch
7) rancidity	7) Do not ~~spoi~~ spoil & turn rancid	7) Can spoil & turn rancid
8) ~~Uses~~ medicinal uses.	8) Antibacterial, antiviral, antifungal, antiparasitic, antiseptic	8) Non Antibacterial, Antiviral, antiseptic, Antiparasic, Antifungal.

Oil V/s Fat

	Fat	oil
1) State at room temp^r	1) Solid at room temp^r	1) Liquid at room temp^r.
2) Sources	2) Obtained from animals i.e. butter, ghee, egg, meat	2) obtained from plant source i.e. mustard, groundnut, almond oil.
3) Fatty acid	3) Fat have saturated fatty acid.	3) oil have unsaturated fatty acid.
4) Bonding	4) No double bond	4) Have double bond.
5) Melting point	5) High	5) low.

Drug V/S Narcotic

	Drug	Narcotic
	~~1) Broad term, effecting the body or mid mind.~~ ~~2) Habit~~	~~1) Specific term, mainly pain relieving~~
1) Purpose	mainly use for medical/ Therapeutic purposes.	mainly use for ~~pe's~~ pain relif.
2) Defin^n	chemical subs. use for diagonosis, prevention or treatment of diseases.	a type of drug that dulls the senses can cause addiction.

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3) Effect on body	can have +ve/−ve effects depending on use and dosage.	prily acts on the CNS to induce sleep or numb sensation.
4) Legal status	Most drugs are ~~le~~legal & prescribed by doctors.	Narcotics are strictly controlled or banned due to their abuse
5) Example	Aspirin	Morphine, Heroin

ϕ What is ethnomedicine?

> The study and practice of traditional healing systems used by different ~~co~~ cultures, focusing on indegenous beliefs about health, disease and treatment, often using local plants and remedies.

ϕ Define Ethnomedicine as per WHO.

> "The sum total of knowledge, skills and practices based on the theories, beliefs, and experiences ~~indog~~ indigenous to different cultures, wheather explicable or not, used in the ~~maintain~~ maintenance of health as well as the prevention, diagnosis, improvement, or treatment of physical and mental illness."

ϕ What is ethnoecology?

> Ethnoecology is an ~~interde~~ interdiciplinary field studying how different human cultures understand, use and interact with their local environment, focusing on Traditional Ecological knowledge (TEK) and its connection to culture, language and survival.

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AEC 1

writting

(For sem1)

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FIR

To,

The Officer-in-Charge
Kotwali Police Station
Coochbehar, West Bengal
Date- 01/02/2026

Subject : Complaint against the thief of my pan card

Respected Sir/ Madam,

I, [redacted], Son of [redacted], resident of [redacted], Coochbehar, wish to lodge a complaint (FIR) regarding an incident that occured on 31/01/2026 at around 8:30 PM.

On the said date, Someone stole my wallet ~~that was in brown colour~~. It at NBSTC Bus Terminus, Coochbehar. The colour of my wallet was brown and it there was around 1500 rupees, including my pan card.

I humbly request you to kindly register my complaint and take necessary legal action. I am ready to cooperate fully with the investigation and provide further information if needed.

Thanking You

Yours Faithfully,
Bikram Singha
MO-

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∅ Job Application

To,
The Manager
TCS Pvt ltd.
Coochbehar

Subject : Job Application

~~Sir~~
Respected Sir/Madam,

I, [redacted], resident of Ghughumari, Coochbehar, came to know ~~through~~ advertisement that many freshers are required for ~~th~~ various position. I consider my eligible for project manager position.

I have done my ~~graduation from ABN Seal College~~ Higher secondary from Virekanda Vidya Peeth, Coochbehar. The resume of my educational Qualifications and ~~int~~ documents are attached with this letter.

Kindly consider my application and give me a chance for interview and examination, I will try my best to satisfy you with my hardwork and complete honesty

Thanking You

Yours faithfully,
[redacted]
MO- [redacted]
[redacted]
Coochbehar
Pin- 736170

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Ø Application for personal loan

To,

The Branch Manager
Punjab National Bank
Gihughumari Branch

Subject: Request for Personal Loan

Respected Sir/Madam,

I, [redacted], resident of [redacted], Coochbehar, maintaining a savings account with your branch for the past 5 five years.

I wish to apply for a personal loan of 2 lakh for the duration of 3 year for my personal use. I have a steady income and a good banking record. Attached are my income proof, ID documents, and completed loan application form.

Kindly consider my request and inform me of the loan eligibility and grant me the loan amount.

Thanking You

Yours Faithfully,
[redacted] Sinha
MO- [redacted]4154
Date- 04/02/2026

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ø Annual Sports day at your College.

Annual Sports Day at ABN Seal College

Coochbehar, 18 December, 2025

—By a staff reporter

The Annual Sports Day of ABN Seal College was held on 17th December 2025 with great enthusiasm and excitement. The college ground was beautifully decorated with colourful flags and balloons. The event began with the hosting of the college flag and a welcome speech by the principal. Students from all departments participated in various events such as 100m races, long Jump, high Jump etc. The professors also took part in the fun games, which added more enjoyment to the day. The SP of Coochbehar district Shri Sandeep Karra sir was the chief guest and he distributed prizes to the winners and prised the students for ~~the~~ their sportsmanship and discipline. The event ended with the National Anthem. Overall the Annual sports day of ABN Seal college was a grand success and a memorable day for everyone.

Same as cultural programe sports replace with Dances, songs, dramas, & instrumental performances.
#Add one more chief guest Subhamita Banerjee.

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Ø Write a report on the Blood donation camp recently held in your college.

Blood Donation Camp in ABN Seal College

Coochbehar, ~~26~~ 27th January, 2026 — By a staff reporter

A Blood Donation Camp was organized in our college, on the occasion of Republic day on 26th January, 2026. The camp was organized by the principal of our college. Two doctors and ten nurses from MJN hospital, Coochbehar, offered their valueable services to the camp. The programme started at 9:00 am in the morning. Many students, ~~teachers~~ professors and college stuffs donated blood with proud hearts. Some of the visitors also donated blood. Before collecting blood everyone taken fruits, juices etc. At around 03:00 PM, Dr. S Banerjee also gave a very good speech on the benifits of Blood donation. A certificate of appreciation was given to each donor as a token of gratitude. Around 400 units of blood was colected and sent to district hospital to ~~f~~ save the precious lives of patients. The programme lasted till 4 PM. This was an act of great services to humanity.

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FIR

To,

The Officer-in-Charge
Kotwali Police Station
Coochbehar, West Bengal
Date - 01/02/2026

Subject - Complaint regarding cyber fraud

Respected Sir/Madam,

I, [redacted], Son of [redacted], resident of [redacted], Coochbehar, wish to lodge a compaint (FIR) regarding an incident that occured on 31/01/2026 at around 8:15 PM.

On the said date, A random link was forworded by someone and I mistakenly opened the link. After clicking that link scammer forworded my bank OTP call on another number and scammed my 20,000 (Twenty Thousand rupees).

My humble request to you to kindly register my complaint and take necessary legal action. I am ready to cooperate fully with the investigation and provide further information if needed. My updated bank passbook The xerox copy of my updated bank passbook attached with this.

Thanking You

Yours Faithfully,
Bikram Singha
Mo- [redacted]

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To,

The Officer-in-Charge
Kotwali Police Station
Coochbehar, West Bengal
Date- 01/02/2026

Subject - Complaint against the thief of Smart phone

Respected Sir/Madam,

I, [redacted], Son of [redacted], Resident of [redacted], Coochbehar, wish to lodge a complaint (FIR) regarding an incident that occured on 31/01/2026 at around 8:30 PM.

On the said date, someone stole my mobile at NBSTC Bus Terminus, Coochbehar. The details of the lost mobile phone as follows:

Mobile company: [redacted]
IMEI~~HEMI~~ NO: [redacted]
Phone NO: [redacted]
Model NO: [redacted]
Mobile Colour: [redacted]

My humble request to you to kindly register my ~~complaint~~ complaint and take necessary legal action. I am ready to cooperate fully with the investigation and provide further information if needed.

Thanking You

Yours Faithfully,
Bikram Singha
Mo- [redacted]

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Geography

PYQ

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2023

1) most abundant element of the earth crust?
> Oxygen.
2) From surface → to centre of earth temperature?
> Increase
3) Pacific Ring of fire –
> Volcano & earthquake.
4) explosive type of volcano-
> Caldera.
5) Quartzite is a-
> metamorphic rock.
6) Sandstone is a – Sedimentary rock
7) Depositional landform at river confluence is ekhane – Delta
8) Not a metamorphic rock- Sandstone
9) Rocks formed by alteration of original structure due to heat and excessive pressure is ekhane – metamorphic rock.
10) Plutonic rock – Granite
11) First recorded wave on the sysmograph- P-wave
12) The scale measureing the earthquake is ekhane – Richter scale.
13) Not correctly matched- Inselberg - Glacier.
14) Triangular shaped deposits formed by transportation of material by water is ekhane - Alluvial fan.
15) Renewable exhaustible resource – Biomass.
16) In an ecosystem, energy flow is always – Unidirectional.
17) Dead Animals → Blowfly maggrot → maggrot → Frogs → snakes
> Detrital food chain.
18) Porosity maximum — clayey
19) Most abundant rock in ~~earth~~ karst region – limestone
20) Increased permeability develops which horizon of soil- A horizon
21) Maximum Biodiversity region of India- Weastern Ghats.
22) Shield is a type of - Plateau.
23) Which is extinct in India – Asiatic cheetah.
24) Arid cycle is associated with – Davis
25) Convection current is found in – Asthenosphere
(পরিচলন স্রোত)

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(*) 26) 'U' shape valley – Glacial (হিমবাহ)
27) Deepest layer of the earth – Core (কেন্দ্র)
28) Permeable rock b/w limestone, marbole & slate – limestone.
29) Which affects rate of infiltration on earth's surface –
> permeability.
30) % of fresh water – (out of total stored H_2O)
> 2.4%
31) Which biome has maximum biodiversity.
> Equatorial Rain forest.
32) What is trophic level in an ecosystem?
> Part of food chain.
33) % of energy transfer from one trophic level to another –
> 10%
34) ~~Selan~~ Solan chak found in –
> Arid region.
35) Transcurrent fault is associated with –
> Transform Boundary.

024

1) Separate crust & the Mantle –
> Moho discontinuity.
2) Highest temp^r layer – Inner core.
3) Rock formed by cooling & solidification of Magma –
> Igneous rock.
(*) 4) Scale use to measure magnitude of earthquake
> Richter scale.
5) Fastest seismic wave –
> P – wave.
6) Landform formed due to deposition of sediments at the mouth of a river –
> Delta.
7) Landform formed due to the accumulation of wind blown sand – Dunes / Sand dunes.
8) Barchan @ is a product of ____ process – Aeolian (বায়ুর)
9) Gir National Park – famous for – Lion

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10) Biodiversity hotspot is- living place for endemic species.
11) Lighthouse of Mediterranean- Stromboli
(ভূমধ্যসাগরীয় বাতিঘর)
12) Plateaus are also c/o- Table land.
13) Highest plateaus in the world- Pamir
14) Transitional zone b/w two biome- ecotone
15) main source of energy in the ecosystem- Sun
16) The main reason of earthquake is- plate movement.
17) The most distructive earthquake wave is- Surface wave.
18) Seismograph was invented by- Milne
19) Geosyncline is- long subsiding region.
20) ~~#~~ ~~#~~ Tethys is an example of- Geosyncline.
21) Fold mountain originates in- Convergent plate boundary.
22) Himalayas originated due to the folding of sediments in the-
> ~~Pa~~ Tethys sea.
23) Fossils are found in-
> Sedimentary (পলল শিলা) rocks.
24) The low lying area b/w two block mountain is c/o
> Rift valley (ভ্রংশ উপত্যকা) (ভূপৃষ্ঠ পর্বত)
25) The Vindhya range is a- Block mountain.
26) Stalactite and stalagmites are found in- Karst region.
27) Geyser is a hot spring — True.
28) Regur is also k/n as- Black soil.
29) Rainfall measure by- Rain gauge.
30) Which of the following is related to water - Hydrosphere
31) give an example of inland drainage in India- Luni.
32) Energy flow in the ecosystem is- unidirectional.
33) Downword movement of H_2O into soil is k/n as-
> Infiltration.
34) Cause of formation of 'U' shaped valley - Glacial erosion.
35) Trees are related to- Transpiration.

Seismograph
↳ instrument
Richter scale
↳ Magnitude scale

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para rubber

-- Priyangshu Banik

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☐ Manufacturing of Para-Rubber:

☐ Manufacturing process of rubber: Natural rubber is obtained from latex, which is a milky substance produce by rubber yielding plant. Main step of the manufacturing process of rubber is tapping and processing.

☐ Tapping of Para-Rubber: This extraction process, known as tapping, can be broken down into the following steps:

(a) The Incision: A specific knife is used to partially cut through the bark of the Hevea tree. These deep cuts extend to approximately half the circumference of the tree.

(b) Direction and Slope: The cut is designed to slope downward to the right, creating a vertical groove that allows the latex to flow efficiently.

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© <u>collection setup:</u> At the lower end of this vertical groove, an iron or zinc spout is fixed. An earthen pot is placed below this incision to collect the dripping latex.

ⓓ <u>Timing of collection:</u> Tapping is primarily done in the morning. This is because turgor pressure is highest at this time, causing the latex to exude more rapidly.

ⓔ <u>Preservation:</u> To prevent the latex from solidifying prematurely, anticoagulants such as hydrogen peroxide (H_2O_2), or ammonia (NH_3) are added to the collection pot to check coagulation of latex

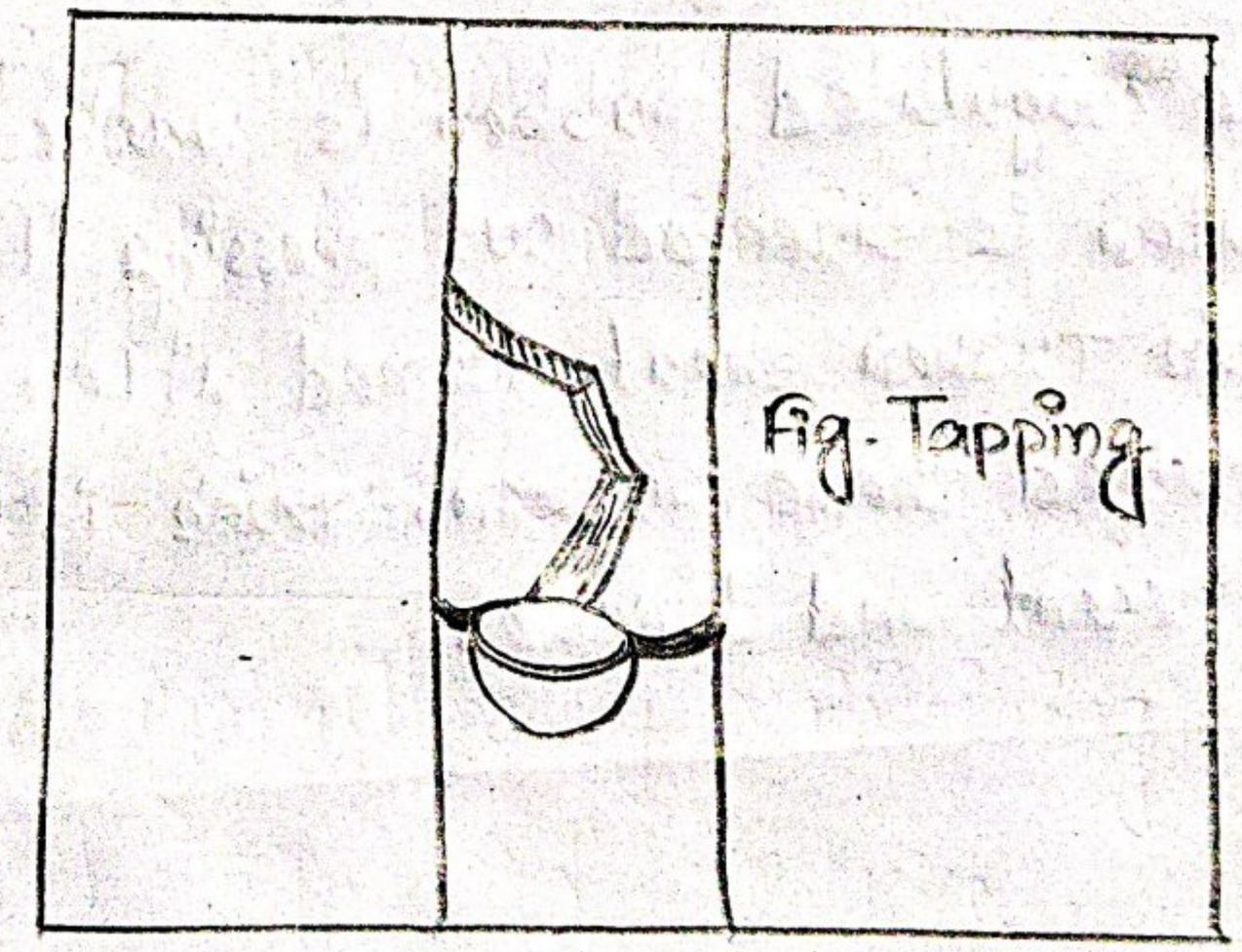


fig- Tapping

□ Processing:

(i) Collected latex is passed through a large aluminium sieve to remove impurites like piece of bark and leaves.

(ii) The water content of the latex is measured with the help of a hygro-meter to standarized, then the latex is poured into a large container where it is allowed to coagulate.

(iii) Coagulation of latex is done by continous stirring with addition of suitable amount of formic acid and acitic acid and some anti-fungals like formiphen, Paranitro-phenol and dinaction and some chemicals are also added to latex to check the growth of fungi and lighten the colour.

(iv) The coagulated rubber is washed with and then water is squeezed out pressing it through roller. The thin rubber sheat formed after pressing the latex is called crepe rubber, which is brown, elastic, translucent and durable.



Fig - Crepe Rubber.

(iv) Vulcanization: This is a chemical process which is required to increased it's tensile, strength, elasticity and resistance. It is done withe temperature at about 150°C with a addition of suitable amount of activation acelearation like sulpher, selenium and Nitrogen. This process involves the formation of cross-link between long rubber molecules to achieve viscosity, hardness and weather resistant.

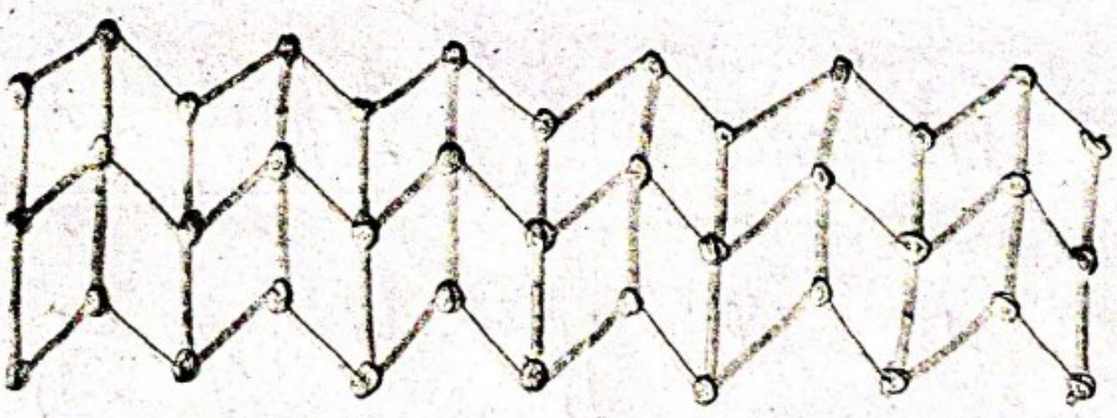


Fig - Cross Link

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THANK YOU

-- Bikram Singha

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